Internet Crimes and Cyber Security

Week: 04 **Session: 19**

Topic :Introduction to Cyber Threats & Attacks

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Cyber Threats & Attacks : Case studies

Concepts to be covered

1. Introduction to Cyber Threats
2. Cyber Threat Maps, Threat Sources.
3. Understand major cyber security threats of 2025
4. What are Cyber Attacks , Classification of Cyber Attacks
5. Analyze real-world cyber attack case studies
6. Identify impacts and preventive strategies

Cyber Threats & Attacks : Case studies

1. Introduction to Cyber Threats

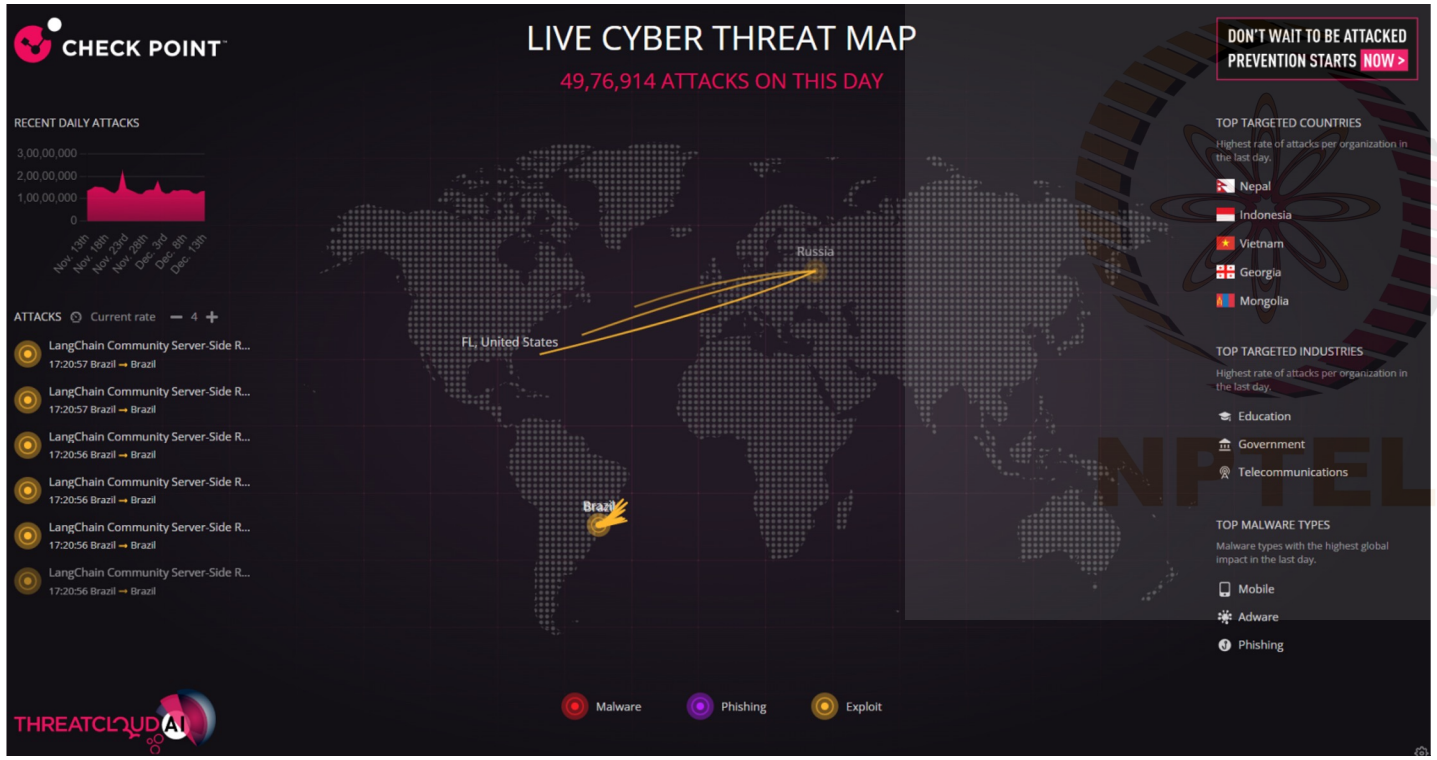
Cyber threats are malicious acts aimed at stealing, damaging , or disrupting data and digital operations, posing significant r sks to individuals and organizations.

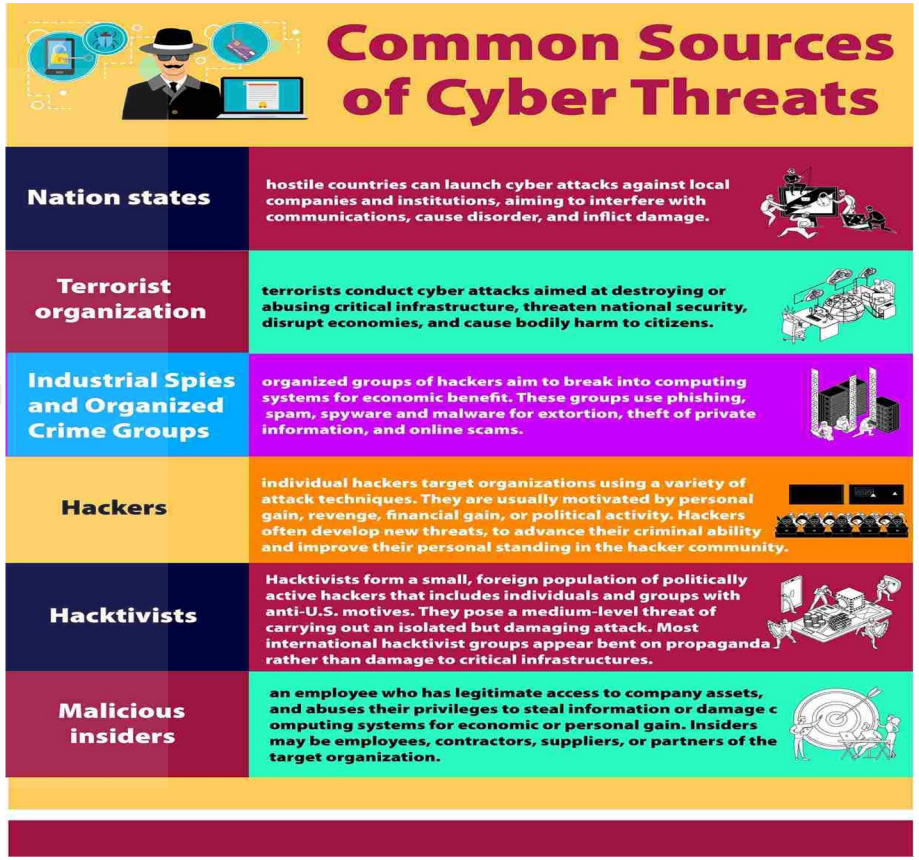
A **cyber threat** refers to a potential danger or risk that could exploit vulnerabilities in a system or asset. It represents the possibility of harm, whether intentional or accidental. Threats can originate from various sources, such as human negligence, natural disasters, or malicious actors. For example,

malware, phishing attempts, or insider threats are all categorized as cyber threats. Importantly, a threat does not necessarily result in harm but indicates the potential for it.

Cyber Threats & Attacks : Case studies

2. Cyber Threat Maps and Threats Sources





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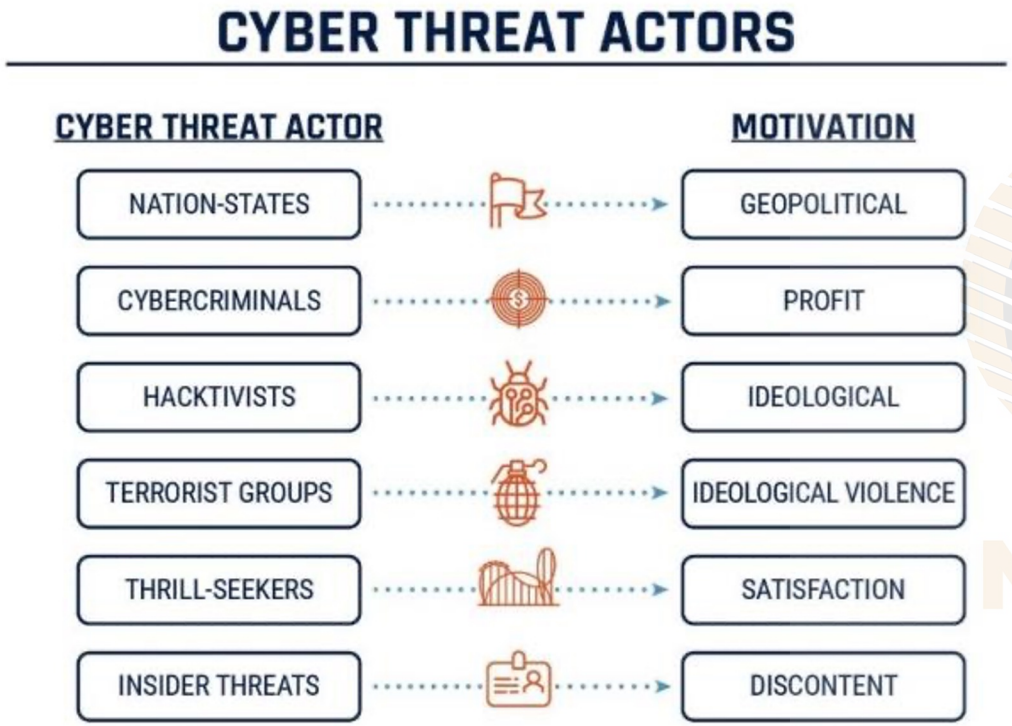
3. Understand major cyber security threats of 2025



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4. What are Cyber Attacks , Classification of Cyber Attacks

An **attack event**, is the actual execution of a malicious action aimed at exploiting a system's vulnerabilities. It is an intentional act carried out by a threat actor to cause harm, steal data, disrupt services, or gain unauthorized access. These Attack events are deliberate and always malicious, such as a Distributed Denial of Service (DDoS) attack, ransomware deployment, or data breaches.

Key differences between Cyber Threat and Cyber attack are :

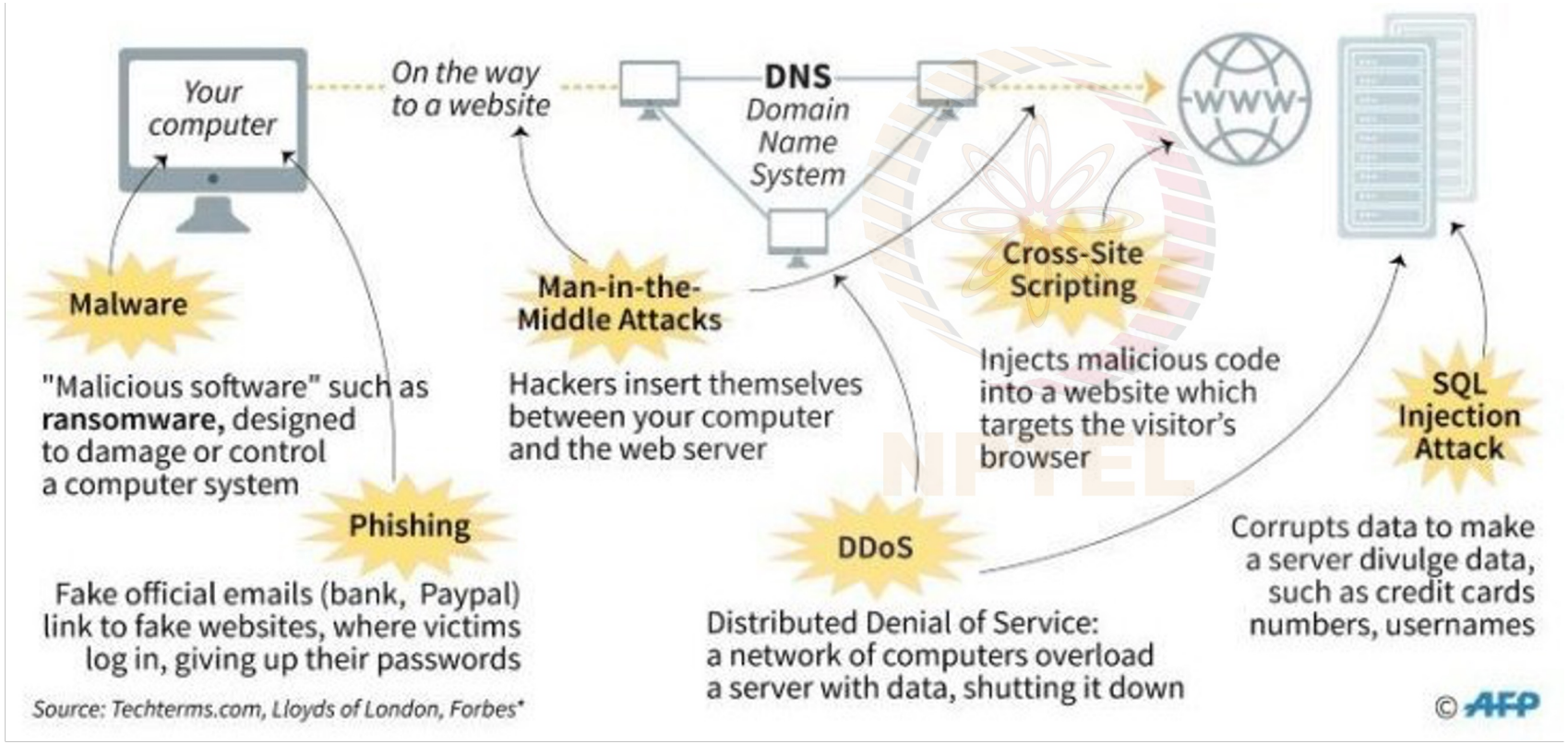
Intent: Threats may or may not be intentional, while attacks are always deliberate and malicious.
Action: A threat is a potential risk, whereas an attack is the realization of that risk through action.
Detection: Threats are harder to detect as they represent possibilities, while attacks are comparatively easier to identify due to their active nature.
Prevention: Threats can often be mitigated by addressing vulnerabilities, but attacks require additional measures like incident response and recovery.

CYBER THREATS
Can be anything from phishing emails to malware
May not be specifically targeted at any one individual or organization
May not result in any damage or harm

CYBER ATTACKS
Are always carried out with malicious intent
Always target a specific individual or organization
Seek to cause damage or disruption

Cyber Threats & Attacks : Case studies

Classification of Cyber Attacks



Thank you